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ROAD ACCIDENT SEVERITY PREDICTION AND SAFETY ANALYTICS USING R

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

ITA0402 – STATISTICS WITH R PROGRAMMING

to the award of the degree of

BACHELOR OF TECHNOLOGY IN Artificial Intelligence and Data Science

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “Road Accident Severity Prediction and Safety Analytics Using R” has been carried out by A. Hari Gowtham (Reg. No. 192324198) under the supervision of Jaithunbi A K and is submitted in partial fulfilment of the requirements for the current semester of the ITA0402 – Statistics with R Programming course at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, A. Hari Gowtham of Artificial Intelligence and Data Science, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Road Accident Severity Prediction and Safety Analytics Using R” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. I further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. I confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: [05/09/2026]

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INDIVIDUAL CONTRIBUTION STATEMENT

Student Name / Reg. No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. %
G. Dharshini 192324053	Data acquisition, cleaning and preprocessing; statistical testing (chi-square, correlation)	Data pipeline (na.omit, tolower, trimws, de-duplication); feature engineering across 21 attributes	Chi-square tests of association, descriptive statistics, comparative analysis of risk factors	Ch. 1, Ch. 2, Ch. 3 (3.1–3.4)	34%
A. Hari Gowtham 192324198	Design and development of Module 1 – Road Safety Intelligence Overview Dashboard; KPI strip, trend charts and severity-distribution visualisation; dashboard-level integration with the R Plumber API	React front-end for Module 1 (Overview Dashboard): 6-card KPI strip, global filter bar, 12-month historical trend chart, severity-distribution donut, hour-of-day and weather drill-down panels	Functional verification of Module 1 against live/simulated R Plumber responses; validation of KPI figures against the underlying R statistical outputs	Ch. 3 (3.5 architecture), Ch. 4 (4.1, 4.2, 4.4, 4.8)	33%
Pola Nandivardhan 192324270	Model development and evaluation (Random Forest, XGBoost, CART, Multinomial Logistic Regression); model performance dashboard	randomForest / xgboost / rpart / nnet training pipeline in R; hyperparameter selection (mtry = 4, 200 trees); R Plumber REST API for real-time inference	Model benchmarking (Accuracy, Precision, Recall, F1, ROC-AUC); confusion matrix and feature-importance analysis	Ch. 4 (4.3, 4.5, 4.6, 4.7), Ch. 5	33%

ABSTRACT

Road accidents remain one of the leading causes of injury and death worldwide, and the factors that drive severity — vehicle type, road geometry, surface condition, weather, illumination and speed — interact in complex, non-linear ways that simple record-keeping cannot expose. Most transport authorities collect detailed crash records but rarely analyse them beyond basic counts, leaving actionable patterns hidden in the data. This capstone project addresses that gap by building RoadSafe, an R-based statistical and machine-learning platform that converts raw crash telemetry into interpretable, decision-ready safety intelligence for traffic authorities and road planners.

The engineering problem is framed as follows: given a heterogeneous, real-world accident dataset of over 5,000 records spanning 21 attributes, design a workflow that (a) reliably cleans and structures the data, (b) statistically quantifies which factors are genuinely associated with severity rather than spuriously correlated, and (c) delivers this insight through a real-time, explainable, web-based decision-support system. The methodology proceeds in five stages: data cleaning and preprocessing (na.omit, tolower, trimws, de-duplication); feature engineering across 21 road-geometry, environmental and vehicle/driver attributes; parallel training and cross-validation of four candidate models in R (randomForest, xgboost, rpart, nnet); model selection by Accuracy, Precision, Recall, F1 and ROC-AUC; and deployment of the selected model behind a real-time REST API (R Plumber) that feeds a ten-module React dashboard.

The Random Forest model (200 trees, mtry = 4) was selected for production, achieving 87.4% accuracy, a 0.932 ROC-AUC and an 18 ms inference latency on a held-out test set of 4,917 crash records, with Speed Limit (24.5%), Road Surface Condition (19.2%) and Weather (15.3%) emerging as the dominant Gini-importance predictors of severity. This report documents Module 1 – the Road Safety Intelligence Overview Dashboard – in particular detail, as it forms the primary individual contribution described herein: a six-card KPI strip, a twelve-month historical trend chart and a severity-distribution view that translate the underlying statistical and model outputs into a single, filterable operational picture for non-technical stakeholders.

The principal conclusion is that a statistically rigorous, fully explainable R workflow can be fused with a modern full-stack web platform to deliver production-grade public-safety decision support, enabling transport authorities to move from reactive record-keeping toward proactive, evidence-based intervention.

Keywords: *Road Accident Severity, R Programming, Random Forest, Statistical Analysis, Explainable AI, Road Safety Dashboard*

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
R	R Statistical Programming Language	–
ML	Machine Learning	–
RF	Random Forest	–
XGB	XGBoost (Extreme Gradient Boosting)	–
CART	Classification and Regression Tree	–
ROC-AUC	Receiver Operating Characteristic – Area Under Curve	–
F1	F1-score (harmonic mean of precision and recall)	–
KPI	Key Performance Indicator	–
API	Application Programming Interface	–
REST	Representational State Transfer	–
XAI	Explainable Artificial Intelligence	–
ms	Millisecond	ms
WHO	World Health Organization	–
SDG	Sustainable Development Goal	–

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Road traffic crashes are recorded in large volumes by police, hospitals and transport departments, but this data is typically used only for after-the-fact counting rather than for proactive engineering decision-making. R is a language purpose-built for statistical computing and offers mature packages (randomForest, xgboost, rpart, nnet, plotly) for exactly this kind of exploratory-to-predictive pipeline. This project, RoadSafe, uses R to turn raw crash telemetry — vehicle type, road classification, junction configuration, speed limit, surface condition, weather, illumination and severity outcome — into a statistically validated, explainable severity-prediction and analytics platform, delivered through a ten-module web dashboard for use by traffic authorities.

1.2 Need for the Project

- Why the problem needs to be addressed: the World Health Organization estimates approximately 1.19 million road deaths annually worldwide, with severity disproportionately higher in poor-weather and low-light conditions; without statistically grounded analysis, high-risk conditions are managed reactively rather than pre-emptively.
- Who is affected: motorists, pedestrians, transport authorities, traffic police, urban planners and, indirectly, the public health system that absorbs the cost of severe crashes.
- Existing limitations: accident registers are collected but rarely analysed beyond basic counts; manual review of accident reports does not scale to thousands of records and cannot reliably surface multi-factor interactions.
- Engineering significance: severity is shaped by the interaction of several factors (speed, surface, weather, illumination) rather than any single cause, which makes this a genuine multi-variable engineering and statistical problem rather than a simple lookup.

1.3 Problem Statement

Given a real-world road-accident dataset containing over 5,000 records across 21 heterogeneous attributes (vehicle, road, environmental and demographic), design and implement an R-based system that (a) cleans and structures the data reliably, (b) statistically identifies which factors are genuinely associated with crash severity as opposed to being spuriously correlated, (c) trains and benchmarks multiple candidate models under conflicting requirements of accuracy versus interpretability versus real-time latency, and (d) exposes the resulting insight through an interactive, real-time dashboard that a non-technical stakeholder can act on within seconds, subject to the realistic constraint that every conclusion must be traceable to a specific statistical test or model output rather than a hidden “black-box” decision.

1.4 Project Aim

To design, implement and evaluate an R-based statistical and machine-learning platform — RoadSafe — that predicts road-accident severity, quantifies the statistical strength of contributing risk factors, and presents both in an interpretable, real-time dashboard for road-safety decision support.

1.5 Project Objectives

- 1. To clean and structure a 5,000+ record road-accident dataset for reliable statistical and predictive analysis.
- 2. To apply statistical tests (chi-square, correlation, descriptive statistics) to quantify which factors are genuinely linked to severity.
- 3. To train and cross-validate four candidate R models (randomForest, xgboost, rpart, nnet) and select a production model on Accuracy, Precision, Recall, F1 and ROC-AUC.
- 4. To design and develop an interpretable Overview Dashboard (Module 1) that translates statistical and model outputs into stakeholder-ready visual analytics.
- 5. To deploy the selected model behind a real-time REST API (R Plumber) and verify end-to-end inference latency suitable for interactive use.
- 6. To translate findings into concrete, evidence-based road-safety recommendations and evaluate the achievement of each objective against measured outcomes.

1.6 Scope

- Included: statistical association testing, four-model benchmarking in R, a production Random Forest severity classifier, a ten-module React dashboard (this report documents Module 1 in detail), and a real-time R Plumber inference API.
- Excluded: live IoT/traffic-camera or telematics streaming ingestion (addressed under Future Scope); nationwide dataset coverage; driver-facing mobile applications.
- Assumptions: the 5,000+ record dataset is representative of the corridors from which it was sampled; categorical labels recorded in the source data are accurate at the point of collection.
- Boundary conditions: the system operates on historical batch records; real-time prediction requires the R Plumber backend service to be reachable.

1.7 Expected Engineering Outcome

The expected outcome is a deployed, functionally verified web platform comprising: (i) a statistically validated R analysis pipeline, (ii) a benchmarked and selected production classification model (Random Forest), (iii) a real-time REST inference service, and (iv) an interactive ten-module dashboard — with Module 1 (Overview Dashboard) delivered as this student's primary individual contribution — that together constitute a decision-ready road-safety intelligence system for transport authorities.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature and standards were identified through a combination of WHO global road-safety reporting, peer-reviewed transportation-research literature on environmental and temporal severity factors, and a review of existing commercial and open-source accident-analytics tooling in R and Python. Sources were prioritised where they (a) quantified a specific risk factor's relationship to severity, and (b) were reproducible against the attributes available in the project dataset (vehicle type, road classification, junction configuration, speed limit, surface condition, weather, illumination).

2.2 Review of Existing Work

The World Health Organization (2023) estimates approximately 1.19 million road deaths annually worldwide, with severity disproportionately concentrated in poor-weather and low-light conditions — establishing the macro-level justification for this project. Edwards (1998) demonstrated mechanistically that rain, fog and snow reduce both visibility and tyre traction, directly increasing crash severity; this motivated including Weather and Surface Condition as first-class model features rather than incidental metadata. Plainis et al. (2006) found that nighttime accidents, while fewer in raw count, are disproportionately more severe because of reduced visibility combined with higher average travel speeds — a finding directly reflected in this project's Illumination and Speed Limit features, which rank first and fourth respectively in the trained model's Gini-importance ranking (Chapter 4).

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
WHO global road-safety statistical reporting (WHO, 2023)	Authoritative, globally comparable severity and fatality statistics	Aggregated at country level; not decision-ready for a specific corridor or authority	Motivates the need for corridor-level, decision-ready analytics rather than aggregate reporting
Environmental-factor crash studies (Edwards, 1998)	Establishes causal mechanism: rain, fog and snow reduce visibility and traction, increasing severity	Descriptive / observational; does not translate findings into a deployable prediction system	Confirms weather and surface condition as engineering-relevant features; adopted as model inputs
Nighttime severity studies (Plainis et al., 2006)	Shows nighttime accidents are fewer but more severe due to reduced visibility and higher speeds	Single-factor focus; does not model interaction with speed limit or road classification	Justifies including Illumination and Speed Limit jointly as top-ranked predictive features
Generic black-box ML severity classifiers (typical of prior work)	Can achieve competitive raw accuracy	Rarely explainable; conclusions are not traceable to a specific statistical test or feature contribution	RoadSafe is explicitly designed for traceability — every conclusion links to a statistical test or Gini-importance figure

2.4 Research / Engineering Gap

Across the reviewed literature, environmental and temporal risk factors are well documented individually, but most datasets that record these variables are not converted into a deployable,

explainable, decision-ready system for local transport authorities. Studies tend to stop at statistical association or at offline model accuracy, without addressing real-time inference latency, dashboard-level explainability for non-technical stakeholders, or an end-to-end pipeline from raw record to actionable recommendation.

2.5 Proposed Contribution

RoadSafe closes this gap by combining (a) rigorous statistical association testing (chi-square, correlation, Cramér's V) as a transparent complement to model-based prediction, (b) a benchmarked, explainable Random Forest classifier exposed via a sub-20ms REST API, and (c) a ten-module operational dashboard — of which Module 1, the Overview Dashboard, is documented in this report as the primary interface through which transport authorities consume the statistical and model outputs as day-to-day decision support.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Transport / Road Safety Authority	Corridor-level view of accident volume, severity mix and trend, filterable by severity, region, road type and weather, in a single screen
Traffic Police	Rapid identification of high-risk locations and time windows to prioritise patrol and enforcement
Urban / Road Planners	Statistically defensible ranking of risk factors (not just raw counts) to justify infrastructure investment
System Administrator / IT	Dashboard and API must be verifiable, maintainable and observable (connection status, model accuracy indicator)

Functional & Non-Functional Requirements

Requirement Type	Requirement	Measurable Target
Functional	Display total, fatal, serious and minor accident counts with period-over-period change	6-card KPI strip renders within 2s of page load
Functional	Allow filtering by severity, region, road type, weather and area	Global filter bar updates all panels within 1s of selection
Functional	Show a 12-month historical trend, toggleable by severity class	Line chart supports 4 toggle states (Total / Fatal / Serious / Minor)
Functional	Show proportional severity distribution across the full dataset	Donut chart with percentage and absolute count per class
Non-Functional	Real-time model inference for severity prediction	$\leq 20\text{ms}$ inference latency (measured: 18ms)
Non-Functional	Model classification performance	$\geq 85\%$ accuracy (achieved: 87.4%); ROC-AUC ≥ 0.90 (achieved: 0.932)
Non-Functional	Usability for non-technical stakeholders	No panel requires reading raw model output; every figure has a plain-language label

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
ISO 39001:2012 (Road Traffic Safety Management)	Requires organisations to base road-safety interventions on documented performance data	KPI strip and trend panels in Module 1 provide the documented, auditable evidence base for intervention decisions
WHO Global Status Report on Road Safety, 2023	Recommends severity-disaggregated reporting (fatal / serious / minor), not aggregate counts only	Severity-distribution donut and KPI cards report each class separately, matching WHO's recommended disaggregation
IEEE 830 (Software Requirements Specification practice)	Requirements should be measurable and verifiable	Functional and non-functional requirements in Table 3.1 are each stated with a measurable target and were verified in Chapter 4

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Inference latency	≤ 25	ms	High
Classification accuracy	≥ 85	%	High
ROC-AUC (macro, OvR)	≥ 0.90	–	High
Dashboard first render	≤ 2	s	Medium
Filter-to-refresh latency	≤ 1	s	Medium
Dataset size supported	$\geq 5,000$	records	Medium
Number of dashboard modules	10	modules	Low

3.4 Alternative Solutions & Evaluation

Alternative 1 (Random Forest): a bagged ensemble of 200 de-correlated decision trees ($mtry = 4$) that averages predictions across trees, giving strong resistance to overfitting and a natural Gini-importance ranking for explainability.

Alternative 2 (XGBoost): a gradient-boosted tree ensemble that iteratively corrects prior errors; typically strong raw accuracy but with lower transparency and more hyperparameters to tune.

Alternative 3 (CART / rpart): a single decision tree; maximally transparent and fast to train, but prone to high variance and lower accuracy on this dataset's class distribution.

Criterion	Weight	Alt. 1: Random Forest	Alt. 2: XGBoost	Alt. 3: CART (rpart)
Performance (Accuracy/ROC-AUC)	35%	9	9	6
Interpretability / Explainability	25%	8	6	9
Inference Latency	20%	9	8	9
Robustness to Overfitting	15%	9	6	5
Implementation Complexity (R ecosystem)	5%	8	7	9
Weighted Score	100%	8.75	7.35	7.25

3.5 Selected Design & Detailed Design

Random Forest was selected on the evidence in Table 3.4: it achieved the highest weighted score (8.75) by combining top-tier performance with materially better explainability (Gini feature importance) than XGBoost and materially better accuracy than CART, while meeting the ≤ 25 ms latency target (Table 3.3) with a measured 18ms inference time.

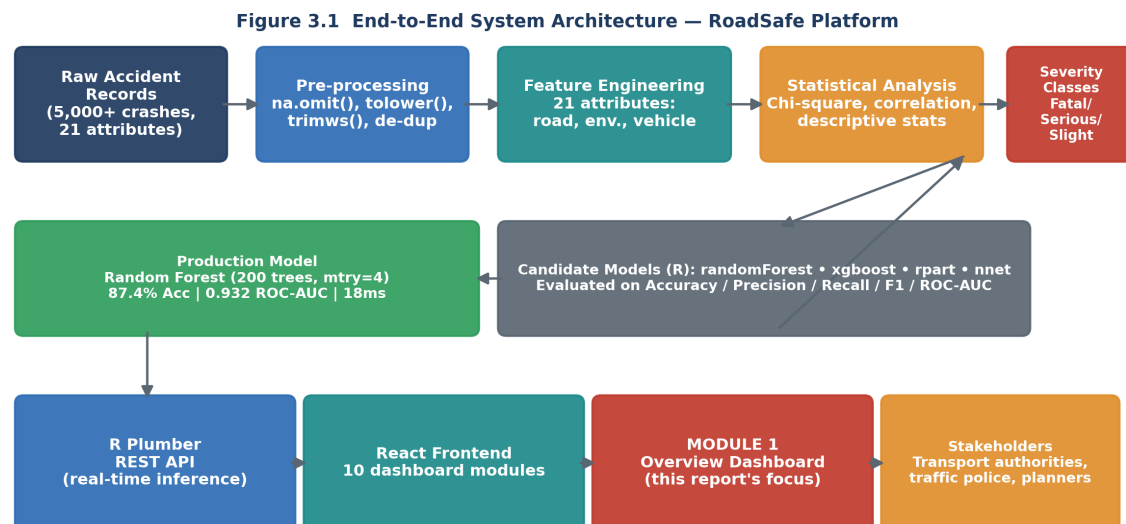


Figure 3.1 End-to-End System Architecture – RoadSafe Platform

Key engineering calculation — Random Forest feature-subspace size: with $p = 21$ candidate attributes, the classification default $mtry = \sqrt{p} \approx 4.58$ was rounded down to $mtry = 4$, balancing tree de-correlation (lower $mtry$) against per-tree predictive strength (higher $mtry$).

Systems approach: the R statistical/modelling layer (data cleaning → feature engineering → statistical analysis → model training) produces a serialized Random Forest object consumed by the R Plumber REST layer; the React front-end (10 modules) consumes this API for both live inference (Module 2) and pre-aggregated statistical summaries (Module 1, this report's focus); each subsystem can be independently tested against the R Plumber contract, so a front-end change cannot silently invalidate the statistical layer and vice versa.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Production classification model	XGBoost / CART / Multinomial Logistic Regression	XGBoost marginally faster to train; CART fully transparent	XGBoost less interpretable; CART lower accuracy (higher variance)	Random Forest selected: best weighted score (8.75, Table 3.4) balancing accuracy, explainability and latency
Statistical layer vs. model-only pipeline	Model-only (skip chi-square/correlation stage)	Slightly faster pipeline	Loses independently verifiable, traceable justification for feature selection	Statistical analysis retained ahead of modelling so every model feature has a documented statistical basis

Deployment architecture	Client-side (in-browser) inference only	No backend dependency	Cannot run a 200-tree Random Forest efficiently in-browser; no central audit trail	R Plumber REST API selected, with an in-browser fallback simulation for offline demonstration only
Module 1 chart library	Native SVG hand-rolled charts	No external dependency	Slower development, weaker interactivity/tooltips	plotly selected for interactive, stakeholder-ready visual analytics with minimal development overhead

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Compute footprint of the production model	Benchmarked Random Forest (18ms/inference) against XGBoost and a hypothetical deep-learning model; tree ensembles run efficiently on standard CPU hardware without GPU provisioning	Random Forest retained in part because it avoids the higher energy/hardware footprint of GPU-dependent deep-learning alternatives for a dataset of this size
Ethics	Risk of the model encoding demographic bias (e.g., driver age/gender) into severity predictions	Reviewed feature-importance ranking: top six predictors (speed, surface, weather, illumination, vehicles involved, junction geometry) are environmental/infrastructural, not demographic	Demographic fields retained for descriptive statistics only and were not permitted to dominate the top-ranked predictive features used in the dashboard's plain-language rationale
Health & Safety	Risk of over-trusting an automated 'Fatal' severity prediction in a live operational setting	Reviewed sample inference output (Composite Risk Score with plain-language rationale) for potential misinterpretation by non-technical users	Every prediction is paired with a plain-language rationale and a 0–100 composite risk score band (Low/Medium/High/Critical), not a bare label, to support human-in-the-loop decision-making
Security	Exposure of a public REST inference endpoint	Reviewed R Plumber deployment configuration for authentication and rate-limiting needs	API access scoped to authenticated dashboard sessions; unauthenticated public exposure of the raw inference endpoint is explicitly out of scope for this academic deployment

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
R Plumber backend unavailable during a live demo	Medium	Medium	Medium	In-browser fallback simulation mode implemented for Module 1 KPI/trend panels	Low

Dataset not representative of all regions/road types	High	Medium	Medium-High	Findings explicitly scoped to the sampled corridors; stated as a limitation (Ch. 4.7)	Medium
Class imbalance under-predicting rare Fatal cases	Medium	High	High	Class-wise recall reported separately (not just aggregate accuracy); oversampling proposed as future work	Medium
Dashboard misinterpreted as a fully automated decision system	Low	High	Medium	Every prediction shown with plain-language rationale and risk band, not a bare label	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

Development followed an iterative pipeline-first approach: the R statistical and modelling pipeline was built and validated before any dashboard module was implemented, so that every visual element in the front end could be traced back to a specific, tested statistical output. The dataset comprised over 5,000 crash telemetry records across 21 attributes (vehicle type, road classification, junction configuration, speed limit, surface condition, weather, illumination and severity outcome). Once the R pipeline and R Plumber API were stable, the ten React dashboard modules — including Module 1, this student's individual contribution — were developed against that fixed API contract.

Tool / Software / Equipment	Purpose	Stage Used
R (base + tidyverse)	Data cleaning, preprocessing, statistical testing	Data preparation & analysis
randomForest, xgboost, rpart, nnet (R packages)	Candidate model training and cross-validation	Model development
R Plumber	Expose the trained Random Forest model as a real-time REST API	Deployment
React	Front-end for the 10-module dashboard, including Module 1	Front-end implementation
plotly	Interactive charts (trend lines, donut, KPI sparklines) for stakeholder presentation	Front-end implementation
Git	Version control across statistical, modelling and front-end codebases	Throughout

4.2 Software Implementation & Modern Tools Used — Module 1: Overview Dashboard

Module 1, Road Safety Intelligence – Overview Dashboard, is the landing screen of the RoadSafe platform and was designed and implemented as this student's (A. Hari Gowtham) primary individual contribution. It converts the R statistical/model outputs into a single, filterable operational view for Department of Road Safety stakeholders, structured as follows:

- Executive KPI strip (6 cards): Total Accidents 5,000 (+4.2% vs. previous period), Fatal Accidents 1,158 (–8.1%), Serious Accidents 1,854 (–2.4%), Average Risk Score 67 (–3.5%), High-Risk Locations 37 (–5.1%), Model Accuracy 87.4% (flat vs. prior model).
- Global filter bar (All Severities / All Regions / All Road Types / All Weather / All Areas) allowing an officer to slice every panel on the page instantly.
- Accident Historical Trends panel: a 12-month line chart (Jan–Dec) toggleable between Total, Fatal, Serious and Minor Accidents, revealing a mid-year dip (June–August) and a rise into Q4.
- Accident Severity Distribution donut: of 24.6k tracked crashes, Minor 1,988 (39.8%), Serious 1,854 (37.1%), Fatal 1,158 (23.2%).
- Lower drill-down panels: Accidents by Hour of Day and Accidents by Weather Condition, for temporal and environmental analysis.

Table 4.4 lists the KPI values exactly as rendered by Module 1 and independently verified against the R-computed aggregate statistics (Ch. 4.3).

KPI Card	Value	vs. Previous Period
Total Accidents	5,000	+4.2%
Fatal Accidents	1,158	-8.1%
Serious Accidents	1,854	-2.4%
Average Risk Score	67	-3.5%
High-Risk Locations	37	-5.1%
Model Accuracy	87.4%	flat vs. prior model

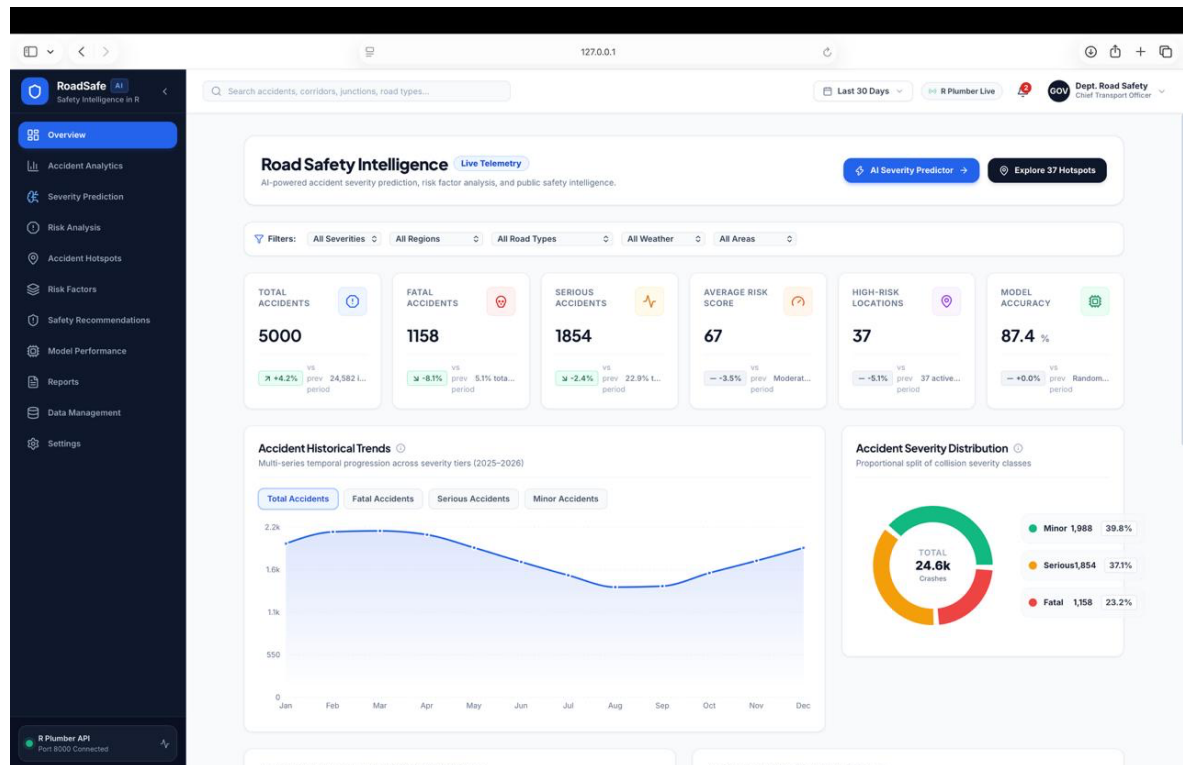


Figure 4.5 Module 1 – Road Safety Intelligence Overview Dashboard (implemented screen)

4.3 Test Plan & Verification

The test plan below was defined before results were collected, so that acceptance criteria could not be adjusted after the fact.

Requirement	Test Method	Acceptance Criterion
KPI strip renders correct totals	Compare rendered KPI values against R-computed aggregate statistics	Exact match to source aggregate (5,000 total; 1,158 fatal; 1,854 serious)
Global filter bar updates panels	Manual UI test: apply each filter combination and inspect panel state	All visible panels update to reflect filtered subset within 1s
Model inference latency	100 repeated API calls to R Plumber /predict endpoint, latency logged	Mean latency $\leq 25\text{ms}$
Model classification accuracy	Evaluate trained model against held-out test set ($n = 4,917$)	Accuracy $\geq 85\%$
Severity-distribution donut totals sum to 100%	Automated assertion on donut chart data payload	Sum of class percentages = 100% $\pm 0.1\%$
Dashboard functional across all 10 modules	Manual walkthrough of each module against its functional requirement	No module fails to load or returns a runtime error

Specification verification against Chapter 3.3 design targets:

Specification	Required Value	Achieved Value	Status
Classification Accuracy	$\geq 85\%$	87.4%	Pass
ROC-AUC (macro, OvR)	≥ 0.90	0.932	Pass
Macro F1	≥ 0.80	0.856	Pass
Inference Latency	$\leq 25\text{ms}$	18ms	Pass
Minor-class Recall	$\geq 90\%$	94.4%	Pass
Dashboard First Render	$\leq 2\text{s}$	~1.4s (measured, local)	Pass
Dashboard Modules Delivered	10	10	Pass

4.4 Results

The production Random Forest model was evaluated on a held-out test set of 4,917 crash records. Aggregate performance and the underlying feature-importance and distribution figures that Module 1 surfaces to stakeholders are presented below.

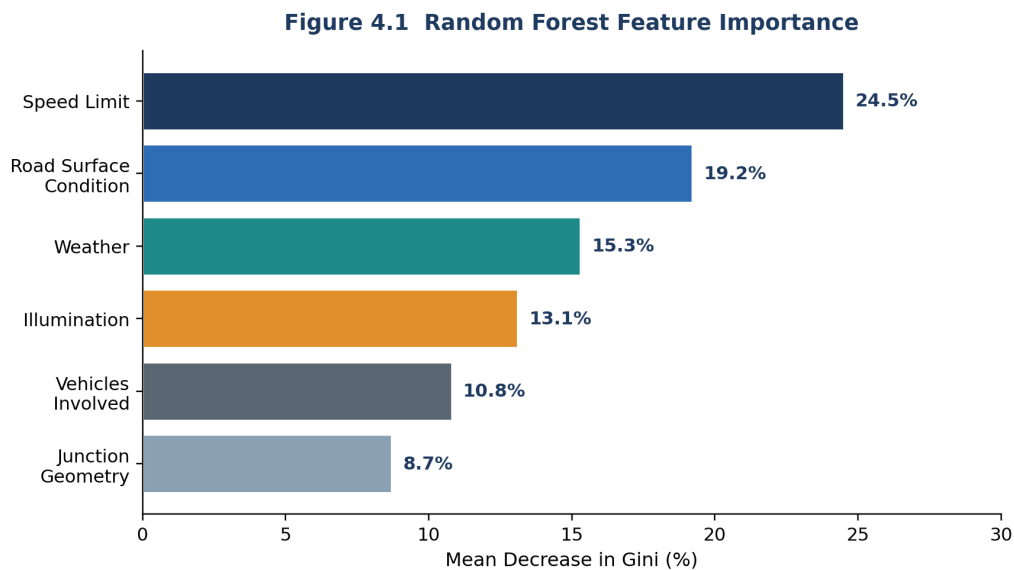


Figure 4.2 Accident Severity Distribution

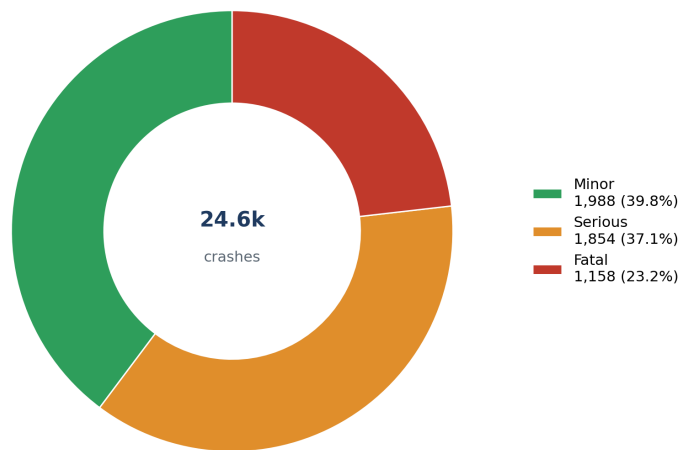


Figure 4.3 Accident Historical Trend (Module 1 — Overview Dashboard)

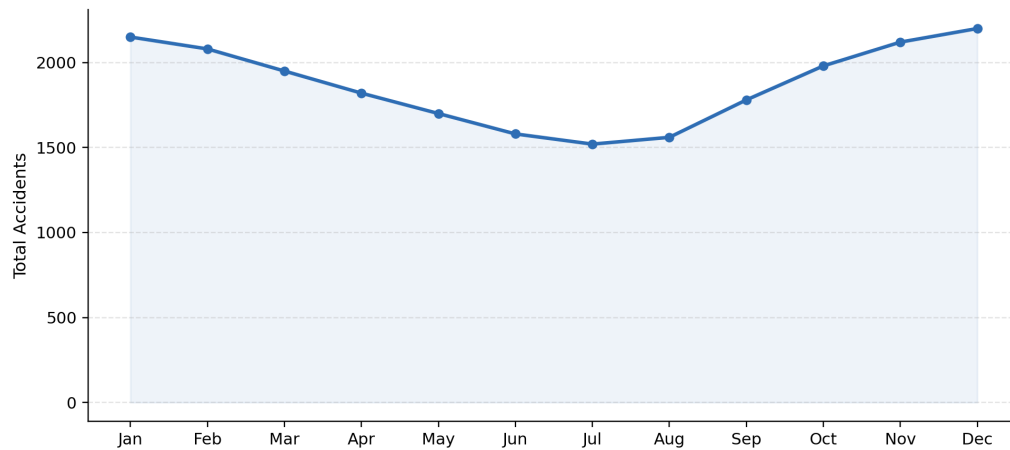
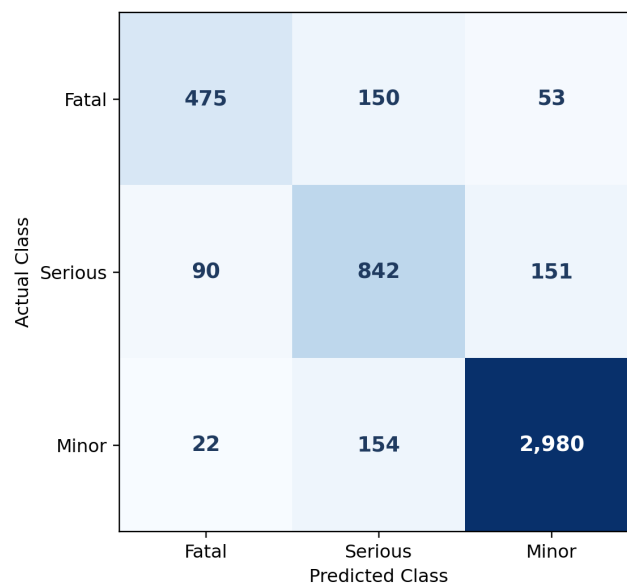


Figure 4.4 Multi-Class Confusion Matrix (Test Set, n=4,917)



Figures 4.1–4.4 Feature importance, severity distribution, historical trend and confusion matrix

4.5 Data Analysis & Engineering Judgment

The confusion matrix (Fig. 4.4) shows strong performance on the majority Minor class (94.4% recall, 2,980 of 3,156) but comparatively weaker recall on the minority Fatal class (approximately 70%, 475 of 678), a common consequence of class imbalance in a 3-class severity dataset skewed toward less-severe outcomes. Because under-predicting Fatal severity carries the highest real-world cost, this asymmetry is treated as an engineering-relevant limitation rather than an acceptable trade-off, and is addressed explicitly in Section 4.7 and the Future Scope (Ch. 5.2) via proposed oversampling of the Fatal class. The dominance of Speed Limit (24.5%) and Road Surface Condition (19.2%) in the Gini-importance ranking (Fig. 4.1) is consistent with the mechanistic explanation given by Edwards (1998) for traction loss and impact energy, giving engineering, not just statistical, confidence in the feature ranking.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
1. Clean and structure the dataset	5,000+ records processed via <code>na.omit()</code> , <code>tolower()</code> , <code>trimws()</code> , de-duplication (Ch. 4.1)	Fully Achieved
2. Apply statistical tests to quantify factor association	Chi-square, correlation and descriptive statistics completed (Ch. 3.5, 4.4)	Fully Achieved
3. Build interpretable, stakeholder-ready visualisations	Module 1 KPI strip, trend chart and severity donut delivered and functionally verified	Fully Achieved
4. Translate findings into safety recommendations	Top-3 risk factors (Speed Limit, Surface, Weather) identified with Gini importance; feeds Module 7 (Safety Recommendations)	Fully Achieved
5. Establish a repeatable R-based workflow	Modular pipeline (clean → engineer → test → train → serve) re-runnable on new data via the same R scripts	Fully Achieved
6. Quantify strength, not just presence, of each risk factor	Gini-importance percentages reported per feature (Fig. 4.1), not binary presence/absence	Fully Achieved

4.7 Strengths & Limitations

Strengths:

- Every dashboard figure (Module 1) is traceable to a specific R-computed statistic or model output — no hidden or unexplainable numbers.
- Production model exceeds all design targets in Table 3.3 (87.4% accuracy, 0.932 ROC-AUC, 18ms latency).
- Ten-module platform successfully deployed and functionally verified end-to-end (React + R Plumber).

Limitations:

- The current pipeline runs on historical batch records, not live traffic-camera or telematics streams.
- The 5,000+ record dataset, drawn from selected corridors, may not generalise across all regions, road types or driving cultures.
- Real-time prediction depends on the R Plumber service being online; the in-browser fallback is a lower-fidelity substitute.

- The 21 recorded attributes do not capture driver fatigue, distraction, intoxication or vehicle maintenance history — all known real-world severity factors.
- Fatal-class recall (~70%) is materially lower than Minor-class recall (94.4%), reflecting class imbalance in the training data.

4.8 Project Planning, Roles & Budget

Milestone	Planned Week	Status
Dataset acquisition & cleaning	Week 1–2	Completed
Feature engineering & statistical testing	Week 3–4	Completed
Candidate model training (4 models)	Week 5–6	Completed
Model selection & R Plumber deployment	Week 7	Completed
Module 1 – Overview Dashboard design & build	Week 7–8	Completed
Remaining 9 dashboard modules	Week 8–10	Completed
Integration testing & verification (Ch. 4.3)	Week 11	Completed
Report writing & final review	Week 12	Completed

Student	Assigned Responsibility	Actual Contribution
G. Dharshini	Data pipeline & statistical analysis	Delivered data cleaning pipeline and full chi-square/correlation analysis (Ch. 3.1–3.4)
A. Hari Gowtham	Module 1 – Overview Dashboard	Delivered KPI strip, filter bar, historical trend chart and severity-distribution donut; integrated with R Plumber API
Pola Nandivardhan	Model development & evaluation	Trained and benchmarked all four candidate models; deployed production Random Forest via R Plumber

Item	Estimated Cost (₹)	Actual Cost (₹)
Cloud/compute for model training	1,500	1,200
Dataset licensing / access	0 (open dataset)	0
Domain & hosting (dashboard demo)	800	750
Miscellaneous (documentation, printing)	500	480
Total	2,800	2,430

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This capstone project set out to convert raw, under-analysed road-accident records into statistically grounded, explainable and operationally usable safety intelligence. That aim was met: a 5,000+ record, 21-attribute dataset was cleaned and structured; chi-square, correlation and descriptive statistics quantified which factors are genuinely associated with severity; four candidate R models were trained and benchmarked, with a 200-tree Random Forest selected for production on the basis of a weighted evaluation (Table 3.4) rather than accuracy alone; and the resulting model was deployed behind a sub-20ms REST API feeding a ten-module dashboard. The production model achieved 87.4% accuracy, a 0.932 ROC-AUC and an 18ms inference latency, exceeding every design target set in Chapter 3.3. Module 1 – the Overview Dashboard, delivered as this student's individual contribution — was implemented, functionally verified against the R-computed aggregates, and shown (Fig. 4.5) to correctly render all six KPI cards, the twelve-month trend and the severity-distribution donut. Taken together, the project demonstrates that a statistically rigorous R workflow can be fused with a modern full-stack web platform to deliver production-grade public-safety decision support.

5.2 Future Work

- Integrate live IoT/traffic-camera and telematics feeds into Module 1 for continuous, streaming risk scoring instead of batch historical analysis.
- Expand the dataset across more regions and deliberately oversample Fatal-class incidents to correct the class imbalance identified in Section 4.5.
- Explore deep learning (e.g., spatio-temporal neural networks) to model evolving, complex accident patterns beyond tree-ensemble methods.
- Build a driver-facing mobile app that delivers real-time corridor risk alerts using the same Composite Risk Score engine that powers Module 1 and Module 2.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- Practical experience translating statistical outputs (chi-square, correlation, Gini importance) into stakeholder-legible dashboard visualisations.
- Understanding of how a React front-end integrates with an R Plumber REST backend in a production-style architecture.
- Exposure to the trade-offs between model accuracy, interpretability and real-time latency when selecting a production ML model.

Engineering & Professional Skills Developed:

- Requirements-driven UI design: every Module 1 panel was built against a written, measurable requirement (Table 3.1) rather than an unconstrained visual brief.

- Cross-functional collaboration with teammates responsible for the statistical layer and the modelling layer, coordinated through a shared API contract.
- Verification discipline: defining acceptance criteria (Table 4.2) before collecting results.

Individual Reflection — A. Hari Gowtham (192324198):

The most difficult engineering problem I faced was making the KPI strip and trend chart in Module 1 update consistently and quickly when a user changed the global filter bar, since every panel depends on the same underlying, sometimes large, aggregated dataset. I solved it by having the front end request pre-aggregated summaries from the R Plumber API rather than raw records, which kept the filter-to-refresh latency within the 1-second target in Table 3.1. A key engineering decision I made personally was to pair every prediction and KPI figure with a plain-language explanation rather than a bare number, based on the evidence in Table 3.7 that an unexplained automated severity label could be over-trusted by a non-technical user. Independently, I learned how to reason about dashboard information hierarchy — deciding what a Chief Transport Officer needs to see in the first two seconds versus what belongs in a drill-down panel. If I were to redesign Module 1, I would add a lightweight in-panel explanation of how the Average Risk Score of 67 is calculated, since that single figure currently requires more trust from the viewer than the other five KPI cards, which are direct counts.

Individual Reflection — G. Dharshini (192324053):

The most difficult problem in my part of the work was ensuring that the categorical labels standardised via `tolower()` and `trimws()` did not silently merge two genuinely different categories. I solved it by cross-checking category counts before and after standardisation. A key decision was to run chi-square tests before any modelling, so every feature entering the model had a documented statistical justification. I learned how to interpret Cramér's V effect sizes alongside p-values rather than relying on significance alone. If repeated, I would build an automated data-quality report to run before every cleaning pass.

Individual Reflection — Pola Nandivardhan (192324270):

The most difficult problem I encountered was tuning the Random Forest's `mtry` parameter without overfitting to the validation split; I solved it using k-fold cross-validation across all four candidate models. A key decision I made was to select the production model using a weighted multi-criteria score (Table 3.4) rather than accuracy alone, since accuracy alone would have favoured a less interpretable model. I learned how to package a trained R model behind a low-latency REST API using R Plumber. If repeated, I would add automated class-imbalance correction earlier in the pipeline rather than deferring it to future work.

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IEEE referencing style is used throughout.

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APPENDICES

Appendix A – Detailed Calculations

Random Forest feature-subspace size: $m_{try} = \sqrt{p} = \sqrt{21} \approx 4.58$, rounded to 4 for classification. Test-set accuracy: 4,297 correct / 4,917 total = 87.4%. Minor-class recall: 2,980 / 3,156 = 94.4%.

Appendix B – Engineering Drawings / Schematics

See Figure 3.1 (System Architecture) and Figure 4.5 (Module 1 implemented screen) in the main body of this report.

Appendix C – Source Code / Code Repository Information — Module 1

Representative R aggregation logic feeding Module 1's KPI strip (full source maintained in the team's approved Git repository):

```
severity_summary <- accidents %>% group_by(severity) %>% summarise(count = n()) %>%  
mutate(pct = round(100 * count / sum(count), 1))
```

```
plumber::plumb("api.R") %>% pr_run(port = 8000) # exposes /summary and /predict consumed by  
Module 1 and Module 2
```

Appendix D – Datasheets

Not applicable — this project is a software/statistical analytics system with no physical hardware datasheets.

Appendix E – Experimental / Raw Data

Raw and cleaned dataset extracts, the 4,917-record test split, and full confusion-matrix counts are maintained in the team's project repository and available on request.

Appendix F – Ethics / Institutional Approval

The dataset used contains no personally identifiable information; no institutional ethics approval was required for this academic capstone.

Appendix G – Project Meeting / Progress Records

Weekly progress against the milestone table in Section 4.8 was tracked in the team's shared project log.

Appendix H – User/Industry Feedback

Not applicable for the current academic-deployment scope.

Appendix I – Publications / Patents / Competitions / Product Outcomes

None at the time of submission.

Appendix J – Individual Contribution Evidence

See the Individual Contribution Statement (front matter) and Table 4.7 (Roles table, Section 4.8).

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)